

Algebra II with Trigonometry

Chapter 6.1

Study guide

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[4506904_alg-2trig-h-chp-6-1-note-docx.pdf](#)

Certainly! Here is a self-contained study guide based on your PDF.

1. Short Summary

This document is an introduction to the unit circle, a central object in trigonometry. It explains the definition and equation of the unit circle, how to locate terminal points based on real numbers, and how to use the reference number (or reference angle) to find coordinates on the circle. The material focuses on understanding how points around the circle relate to real numbers (interpreted as arc length or angle), the symmetry of the circle, and provides several practice problems.

2. Core Definitions, Theorems, and Formulas

Key Definitions

- **Unit Circle:** The circle of radius 1 centered at the origin $(0, 0)$ in the xy -plane.

Equation: $x^2 + y^2 = 1$

- **Terminal Point:** For a real number t , the terminal point $P(x, y)$ is reached by moving a distance t along the unit circle from $(1, 0)$, counterclockwise if t is positive and clockwise if t is negative.

- **Circumference of the Unit Circle:** $C = 2\pi$
- **Reference Number (Reference Angle):** The shortest distance along the unit circle between the terminal point determined by t and the nearest point on the x-axis.

Formulas

- **Quarter Circle:** Distance $\frac{C}{4} = \frac{\pi}{2}$
 - **Half Circle:** Distance $\frac{C}{2} = \pi$
 - **Coordinates on Circle:** If $P(x, y)$ is on the unit circle, then $x^2 + y^2 = 1$.
 - **Reference Number Calculation by Quadrant:**
 - Quadrant I: $\bar{t} = t$
 - Quadrant II: $\bar{t} = \pi - t$
 - Quadrant III: $\bar{t} = t - \pi$
 - Quadrant IV: $\bar{t} = 2\pi - t$
 - **Periodic Property:** The terminal point for t is the same as for $t + 2\pi n$, where n is any integer.
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3. Study Guide: Step-by-Step Explanation

Step 1: The Unit Circle

- Draw a circle with center at $(0, 0)$ and radius 1.
- Its equation: $x^2 + y^2 = 1$.

Step 2: Terminal Points

- Start at $(1, 0)$.
- Move counterclockwise for positive t , clockwise for negative t .
- The circumference is 2π (full turn), so moving a distance t corresponds to an angle t radians.
- Special points:
 - $t = 0$: $(1, 0)$

- $t = \frac{\pi}{2}$: (0, 1)
- $t = \pi$: (-1, 0)
- $t = \frac{3\pi}{2}$: (0, -1)
- $t = 2\pi$: (1, 0)

Step 3: Reference Number

- The reference number helps find the coordinates for any t , using the symmetry of the circle.
- Identify the quadrant where the terminal point lies.
- Compute $\bar{t}$ as per quadrant.

Step 4: Finding Coordinates

- Calculate the reference number $\bar{t}$, whose terminal point $Q(a, b)$ is easier to find.
- The actual point P has the same coordinates with signs chosen by the quadrant.

Step 5: Periodic Property

- If t is outside $[0, 2\pi)$, add or subtract 2π repeatedly until $0 \leq t < 2\pi$.
- The corresponding point on the circle does not change.

4. Practice Problems with Answers

1. Fill in the blanks:

- The unit circle is the circle of ___ radius 1 ___ centered at the ___ origin ___ in the xy-plane. Its equation is $___ x^2 + y^2 = 1 ___$.
- The circumference of the unit circle is $___ 2\pi ___$.

2. Find the reference number for $t = \frac{11\pi}{6}$:

- Since $\frac{11\pi}{6}$ is in Quadrant IV: $\bar{t} = 2\pi - t = 2\pi - \frac{11\pi}{6} = \frac{12\pi}{6} - \frac{11\pi}{6} = \frac{\pi}{6}$.

3. Find the missing coordinate: $P(-\frac{3}{5}, ?)$ lies on the unit circle in Quadrant III.

- Since $x = -\frac{3}{5}$, and in Quadrant III, $y < 0$.
- From $x^2 + y^2 = 1$, solve for y :

$$\left(-\frac{3}{5}\right)^2 + y^2 = 1$$

$$\frac{9}{25} + y^2 = 1$$

$$y^2 = 1 - \frac{9}{25} = \frac{16}{25}$$

$$y = -\frac{4}{5} \text{ (Quadrant III: } y \text{ is negative)}$$

$$\textbf{Answer: } P\left(-\frac{3}{5}, -\frac{4}{5}\right)$$

4. Find the terminal point for $t = -\frac{5\pi}{4}$:

- Bring to $[0, 2\pi)$: add 2π

$$-\frac{5\pi}{4} + 2\pi = -\frac{5\pi}{4} + \frac{8\pi}{4} = \frac{3\pi}{4}$$

- $\frac{3\pi}{4}$ is in Quadrant II. Reference number: $\bar{t} = \pi - t = \pi - \frac{3\pi}{4} = \frac{\pi}{4}$.
- Terminal point for $\frac{\pi}{4}$: $\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$
- Adjust sign for Quadrant II: $\left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$

5. Suppose the point determined by t is $\left(\frac{3}{5}, \frac{4}{5}\right)$. What is the terminal point for $\pi - t$?

- For $\pi - t$, the terminal point is $\left(-\frac{3}{5}, \frac{4}{5}\right)$ (reflect about y-axis).

5. Assumptions and Ambiguities

- The reference number is also known as the reference angle in other textbooks.
- The document uses radians throughout; angle measures in degrees are not shown.
- Signs for coordinates are taken based on the quadrant; knowledge of quadrant signs is assumed.
- Some special terminal points (i.e., for $\frac{\pi}{6}$, $\frac{\pi}{4}$, etc.) are referenced but their coordinates are not given; typical values are assumed.

- Formulas for finding terminal points are assumed to use standard trigonometric coordinates.
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Let me know if you'd like even more detailed worked examples or additional practice!